

Whole-Home Electrification Project

Palo Alto, CA

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About Me

- Silicon Valley systems/software/hardware developer, semi-retired
- Highly technical and hands-on, still writing code
- Home automation enthusiast
- Technology policy interests: energy, electrification, grid modernization

Motivations

- Always wanted rooftop solar — grid-tied seemed incomplete, BESSes and EVs started getting good
- Gen-Z daughter very concerned about sustainability — urged us to "do better"
- Appointed to Utilities Advisory Commission, learned that natural gas is **bad**
- As a strong advocate for electrification, I believe in **practicing what I encourage others to do**

Starting Condition

- Purchased in 2015
- Needed a new roof
- Aging HVAC and water heater
- Kitchen appliances: functional but dated
- Electric service: **200A**, main and sub panels
- Second floor uncomfortably hot in summer, A/C ineffective
- Three ICE vehicles

Goals

- **Replace all gas appliances** with electric, disconnect from gas utility
- **Support EV charging** (potentially 2 EVs)
- **Resilience during grid outages:** rooftop solar (NOT grid-tied) + battery backup
- We plan to live here "forever" — quality of life and future-proofing outweighed budget concerns
- **Demonstrate** how a fully electrified home can actively participate in a dynamic, flexible grid
- Create a **real-world example** of smart load management, storage, and automation
- Become a working **laboratory** for energy and power management based on dynamic price, grid conditions, and household needs

Design Process

- Installed networked multi-circuit electric meter to **measure existing use**
- NEC load calculations are insanely conservative — a major obstacle to electrification
- Concluded whole-home backup better/more future-proof than partial-home
- Configurable & automated load-shedding during outages highly desirable
- Roughly **2+ years** of research and planning

Pre-Electrification Baseline (2020)

Consumption history

Data for Selected Service

General Data

Meter Reads

Consumption History

Consumption History Data (Meter number 41858)												
Year	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
2020	2,940	0	0	0	0	0	0	0	0	0	0	0
2019	2,435	2,622	2,379	2,141	2,067	2,599	2,541	2,549	2,712	2,717	2,473	2,438
2018	3,466	1,165	2,410	2,093	1,972	2,361	2,115	2,248	2,263	1,939	2,445	2,308
2017	2,371	2,609	2,114	2,254	2,130	2,373	2,024	2,240	2,243	1,953	1,839	2,192

The Consumption History dates are based on the last meter read dates.

Utility meter reads

Data for Selected Service

General Data

Meter Reads

Consumption History

Meter Number	Meter Reading Type	Read Date	Meter Reads	Unit
41858	Meter reading by utility company	Feb 6, 2019	35352.0	KWH
41858	Meter reading by utility company	Mar 7, 2019	37731.0	KWH
41858	Meter reading by utility company	Apr 4, 2019	39872.0	KWH
41858	Meter reading by utility company	May 3, 2019	41939.0	KWH
41858	Meter reading by utility company	Jun 5, 2019	44538.0	KWH
41858	Meter reading by utility company	Jul 5, 2019	47079.0	KWH
41858	Meter reading by utility company	Aug 5, 2019	49628.0	KWH
41858	Meter reading by utility company	Sep 5, 2019	52340.0	KWH
41858	Meter reading by utility company	Oct 7, 2019	55057.0	KWH
41858	Meter reading by utility company	Nov 5, 2019	57530.0	KWH
41858	Meter reading by utility company	Dec 4, 2019	59968.0	KWH
41858	Meter reading by utility company	Jan 6, 2020	62908.0	KWH

Average daily

Your Average Daily Consumption and Cost Table

MONTH	SERVICE DAYS	ELECTRIC		GAS		WATER	
		KWHS PER DAY	DOLLARS PER DAY	THERMS PER DAY	DOLLARS PER DAY	GALLONS PER DAY	DOLLARS PER DAY
12/2019	29	84.1	15.69	1.78	2.18	438.48	5.90
11/2019	29	85.3	15.92	0.87	1.36	593.24	7.98
10/2019	32	84.9	15.85	0.53	0.90	607.75	8.18
09/2019	31	87.5	16.35	0.31	0.71	579.10	7.79
08/2019	31	82.2	15.33	0.30	0.71	579.10	7.79
07/2019	30	84.7	15.68	0.42	0.76	598.40	7.68
06/2019	33	78.8	14.50	0.69	0.97	589.33	7.40
05/2019	29	71.3	13.06	0.61	0.96	232.14	2.99

Design Decisions

Component	Choice
Roof	Standing-seam metal, replacing aging Hardie-shake
Insulation	2.5" solid foam under roof + new attic insulation
HP Water Heater	Rheem 50 gal, 240 VAC, 30A
HP HVAC	Lennox variable-speed, multi-zoned
Solar PV	LG 400W panels, Enphase microinverters, rail-less mount
Battery	Tesla Powerwall 2 (three units)
Cooktop	50A induction hob
Panels	Two SPAN smart panels in series
EV + EVSE	Tesla Model Y + Wall Connector (Level 2, 60A)

Before & After: Service Entrance

Before



After



New panels + Tesla Powerwall

SPAN Smart Electric Panels

- Home automation and energy management systems
- Real-time power monitoring of each circuit
- Controllable relay on each circuit
- Configurable automated load-shedding during outages



Before & After: HVAC and Water Heater

Before

Aging gas furnace and gas water heater



After

Rheem HP-WH (50 gal, 240V) + Lennox variable-speed HP-HVAC, multi-zoned



New Roof and PV Panels

- Standing-seam metal roof with S-5! PVKIT rail-less mount
- LG 400W panels with Enphase microinverters
- Conduit from rooftops to service entrance routed through former furnace/WH flue



EV Charging (EVSE)

- Tesla Wall Connector, Level 2, 60A
- Installed in garage for primary EV
- Capacity allocated via SPAN smart panel



Induction Cooktop

Before (gas)



After (induction)



- 50A induction cooktop in same location as gas
- Moderate kitchen remodel: new countertop, backsplash, new appliances, kept cabinets
- Moved ventilation hood into ceiling — no gas fumes to exhaust!

Disconnecting from Gas



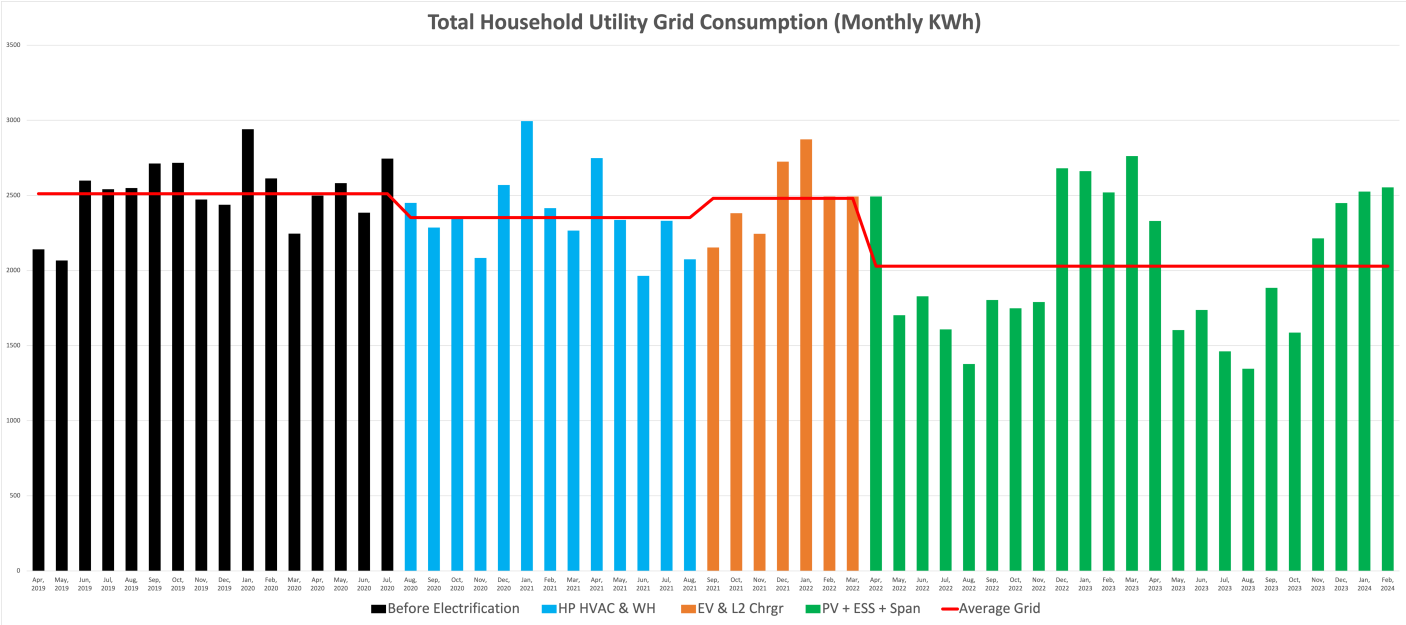
Data: Monthly Grid Consumption

Pre-electrification: ~2000 kWh/month

Post-electrification: ~2500 kWh/month

But grid consumption actually **decreased** — from ~2500 to ~2000 kWh from grid.

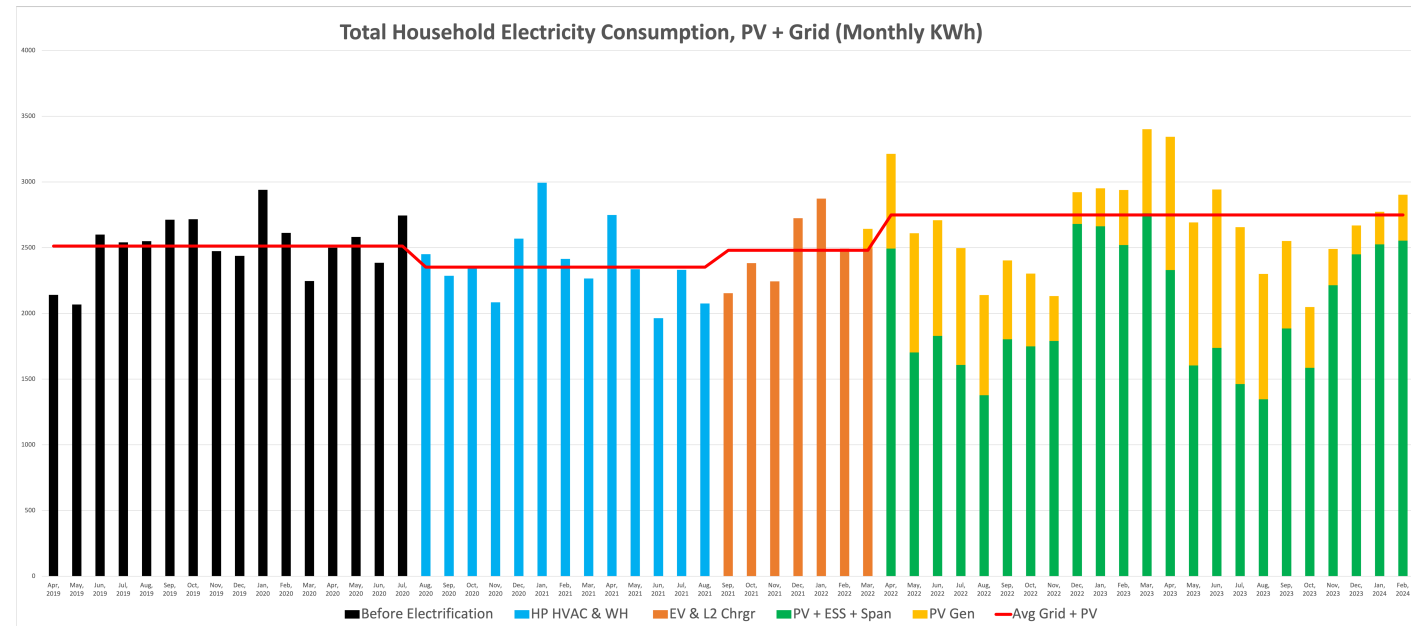
The difference: ~750 kWh/month of PV generation



Data: Monthly Total Consumption (Grid + PV)

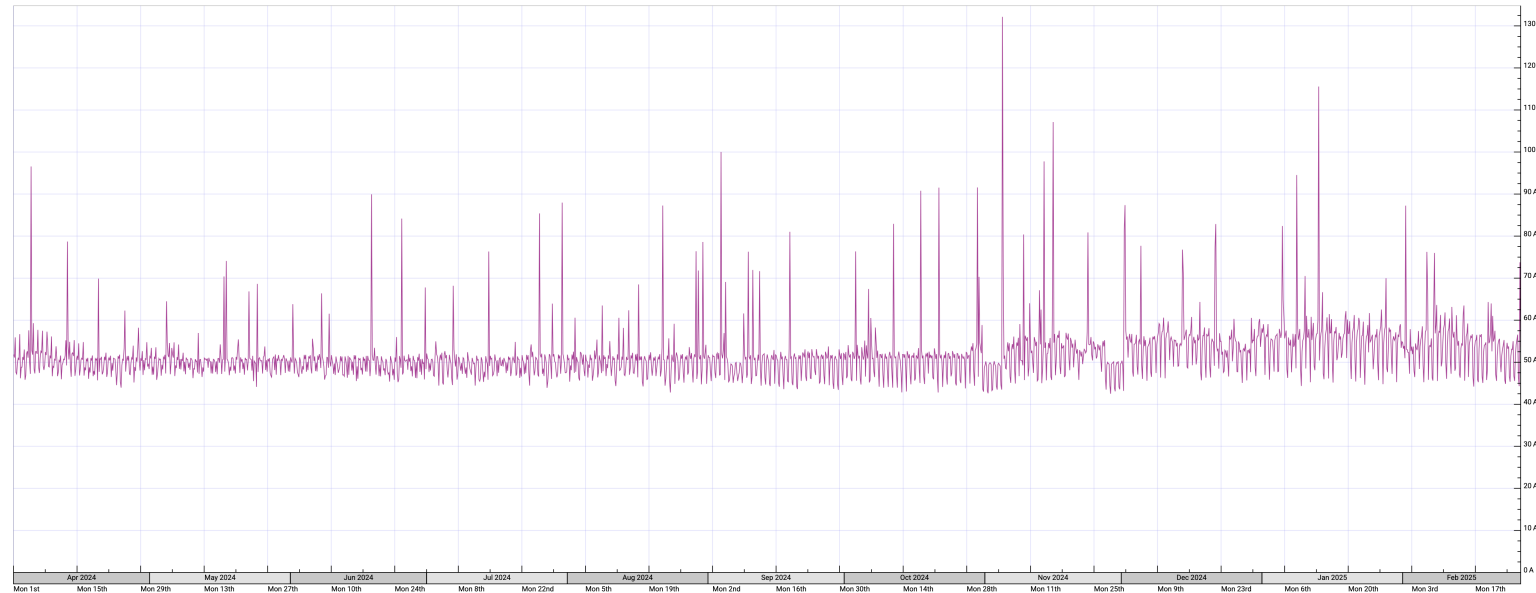
Total consumption: ~2750 kWh/month

- Using ~250 kWh **more** electricity than pre-electrification
- But ~500 kWh **less** from the grid
- PV generation makes up the difference: ~750 kWh



Data: Instantaneous Grid Consumption

- Large baseline load of **~3 kW** with occasional large spikes/peaks
- (Mini "data center" in the basement!)
- Peaks of **15-20 kW** for a few hours — EV charging + cooking or dryer



Data: Peak Usage Event (Zoom)

A typical peak: EV charging overlapped with cooking and dryer.



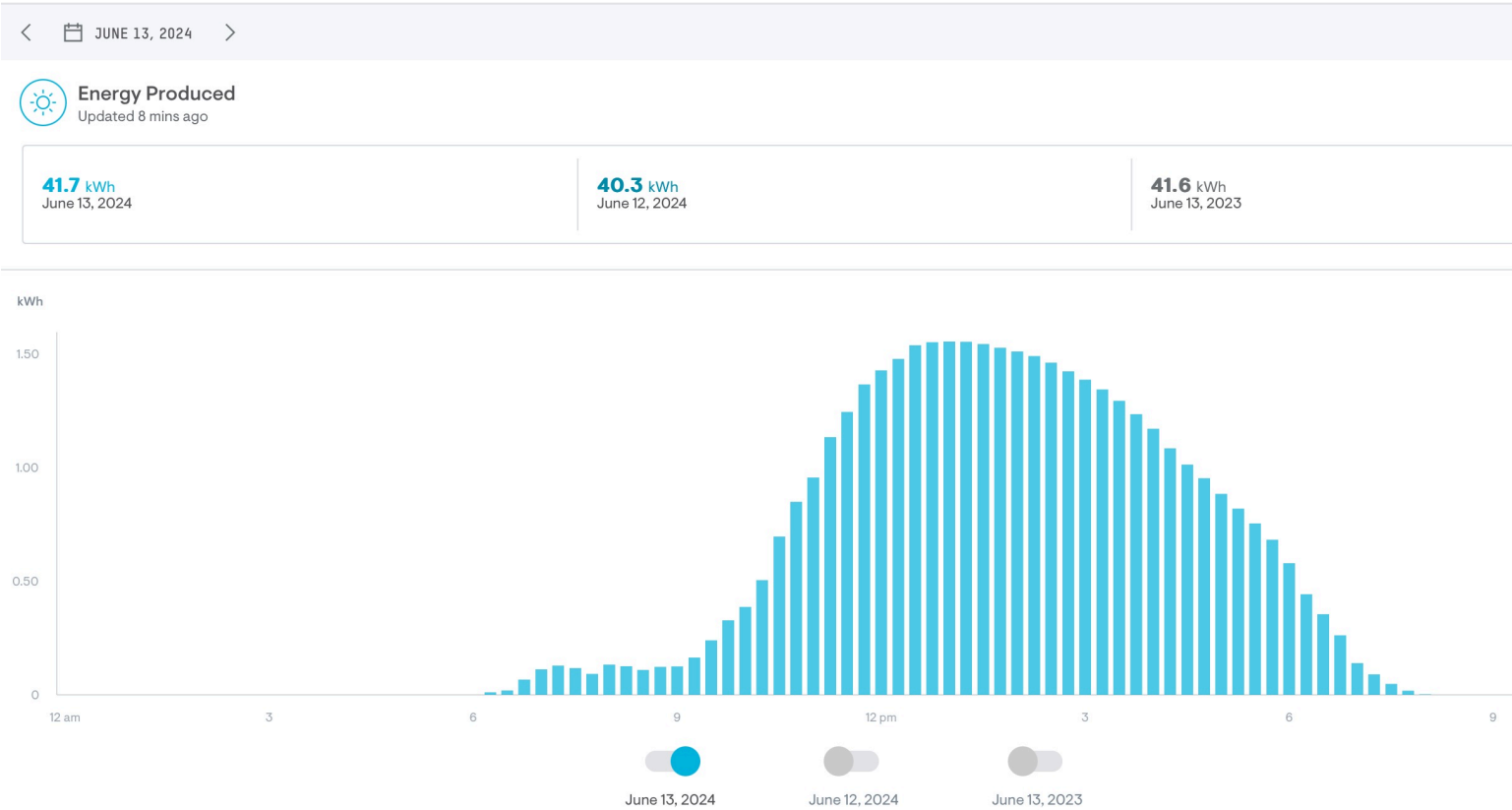
Data: PV Generation 2024 — Monthly

- 8.1 MWh total for 2024 (8.4 MWh in 2023)
- Monthly range: ~0.25 MWh (winter) to ~1.2 MWh (summer)



Data: PV Generation — Best Day (June 13, 2024)

- 41.7 kWh generated on June 13, 2024
- Peak output around solar noon



Conclusions from the Data

- On average, using **250 kWh more** electricity per month than pre-electrification
- But using **500 kWh less** from the grid
- PV generation makes up the difference: **750 kWh**
- **200A service is more than sufficient** — even without active load management
- Additional electricity cost without PV is immaterial
- With PV: completely electrified **and** lowered the electricity bill
- Gained **resilience** to grid outages

Life After Electrification

System	Experience
Insulation	Massive improvement — second floor never more than a couple degrees warmer. Added ~20% to roofing cost, money very well spent
Hot Water	Zero change, no problems. HP-WH always uses heat pump, never resistive coils
HVAC	Vastly improved A/C. Multi-zone variable speed = better comfort. Space heating: does the job. Very efficient and quiet
Induction Cooktop	Just as good, if not better. Electronic temp control arguably more precise
EV	Best vehicle I've ever owned. Level 2 home charging vastly superior to gas stations

No compromises, sacrifices, or negative lifestyle impacts. Electrified life outperforms in every way — plus it's cleaner, safer, and guilt-free.

Service Feed Size

- Based on experience and data, **200A is more than enough** for nearly all homes
- 400A service seems unnecessary — NEC load calcs are insanely conservative
- What about homes with **100A service**?
 - EMS/PCS technology (UL 916 smart panels) can intelligently manage total load
 - Most 100A homes can electrify **without a costly service upgrade**
 - Smart panel investment pays for itself through avoided upgrade costs

| *Existing service feed size is NOT an obstacle to electrifying most homes.*

Grid Impacts of Electrification

- Data suggests grid impact projections may be more conservative than necessary
- Current adoption rates for HP-WH and HP-HVAC have modest grid impact
- No need to hold back motivated customers while waiting for grid upgrades

The opportunity:

- Support customer adoption while upgrading the grid **in parallel**
- Crawl, walk, run — start now and scale gradually
- Rethink grid management: balance responsible management with ongoing electrification

Smarter Grid Management with Smart Controls

Historically, grids were designed to handle worst-case peak load with no active demand management.

Today: EMS/PCS technology in smart panels offers real-time, per-customer load management.

Key question:

How does the cost of installing EMS/PCS panels at every home compare to grid modernization?

Example: 15K homes x \$1K = \$15 million

With granular control, the distribution network no longer needs to be oversized for unconstrained demand. Demand can be optimized in real time — **deferring or avoiding costly infrastructure upgrades.**

Where the Puck Is Going

- **Hourly dynamic pricing** via open protocols like OpenADR 3.1
- **Bi-directional pricing** — addressing NEM concerns, incentivizing storage
- **V2X-enabled EVs** — household backup power and flexible grid resource
- **Fixed connection charges** based on service feed size: \$(100A) < \$(200A) << \$(400A)
- **All high-load appliances** will be network-connected and demand-flexible (CEC FDAS)
- **Home automation** going mainstream via Matter — active energy optimization

"Nobody Else Can Afford This!"

I respectfully disagree. Many people can do most of this, **over time**, with planning:

Component	When
HP Water Heater	When your WH needs replacing (10-12 years)
HP HVAC	When your HVAC needs replacing (15-20 years)
Induction Cooktop	Kitchen update
EV	Plan for your next vehicle to be electric
Battery	Wait for V2X-EVSE + EV — no need for dedicated BESS
Smart Panel	At 100A service: cheaper than a service upgrade
Solar PV	Totally optional — your choice

Electrification is a journey, not a single project.

Surviving a Grid Outage (Electrified)

Need	With Battery Backup	Without
Fridge, lights, Internet	Easy	Same as anyone
Hot water	HP-WH very efficient on battery	50+ gal tank lasts a while
Cooking	Possible but big drain — use outdoor BBQ	Gas cooktops can be manually lit
HVAC	Possible but significant drain	Gas furnace needs electricity too
EV charging	Would drain all capacity — keep EV charged!	N/A
Laundry	Defer until power restored	Same

| *Keep your EV charged. Most gas appliances need electricity too. Battery backup covers essentials.*

Final Thoughts

- Practicing what we preach: fully electrified, lower bills, better comfort, grid resilient
- 200A service is sufficient — smart panels make even 100A work
- Grid impacts are manageable — start electrifying now, upgrade grid in parallel
- Dynamic pricing + smart controls = the future of grid management
- **No compromises.** Electrified life is simply better.